

International Institute of Information Technology, Bhubaneswar

Department of Computer Science and Engineering

Object Oriented Programming Laboratory**Lab 4 – C++ Programming: Friend Function and Friend Class****Date:** 18.08.2026**Semester:** B.Tech 3rd Semester**Section:** CSE B2**Instructions**

- Write all programs in C++.
- Use the **friend function** or **friend class** concept as specified in each question.
- Use appropriate classes, objects, data members, and member functions.
- Display the output in a clear and readable format.
- Keep the required data members **private** and provide access through friendship where specified.

1. Personal Diary

Create a class named `Diary` containing the following private data members:

- Owner Name
- Number of Entries
- Last Entry

Write a friend function named `displayDiary()` that accesses and displays the private details of a `Diary` object.

***Hint:** Declare `displayDiary()` as a friend function inside the `Diary` class.*

2. Mobile Phone Settings

Create a class named `Mobile` containing the following private data members:

- Brand
- Model
- Battery Percentage

Write a friend function named `checkBattery()` that accesses the private battery percentage and displays:

- "Battery Low" if the battery percentage is below 20%.
- "Battery Normal" otherwise.

The function should also display the mobile details.

***Hint:** The friend function can directly access the private members of the `Mobile` class.*

3. Parking Slot

Create a class named `ParkingSlot` containing the following private data members:

- Slot Number
- Vehicle Number
- Occupancy Status

Write a friend function named `checkSlot()` that accesses the private members and displays whether the parking slot is occupied or available. If the slot is occupied, display the vehicle number as well.

Hint: Use a suitable Boolean variable to represent whether the slot is occupied.

4. Music Playlist

Create a class named `Song` containing the following private data members:

- Song Name
- Artist Name
- Duration

Create two `Song` objects. Write a friend function named `compareSongs()` that compares the duration of the two songs and displays which song is longer. If both songs have the same duration, display an appropriate message.

Hint: The friend function should receive both `Song` objects as arguments.

5. Food Order

Create a class named `FoodOrder` containing the following private data members:

- Order ID
- Food Item
- Quantity
- Price

Write a friend function named `calculateBill()` that accesses the private members and calculates the total bill.

Display the complete order details along with the total bill.

Hint: Total Bill = Quantity × Price.

6. Smart Door Lock

Create two classes, `Door` and `SecuritySystem`.

The `Door` class should contain the following private data members:

- Door Number
- Lock Status

The `SecuritySystem` class should contain a member function that checks the lock status of a `Door` object.

Declare `SecuritySystem` as a **friend class** of `Door`.

The program should display whether the door is “Locked” or “Unlocked”.

Hint: Since `SecuritySystem` is a friend class, its member functions can directly access the private members of `Door`.

7. Game Player Status

Create a class named `Player` containing the following private data members:

- Player Name
- Health
- Score
- Level

Create a class named `GameManager` and declare it as a **friend class** of `Player`.

The `GameManager` class should contain member functions to:

1. Display player details.
2. Check whether the player is alive.
3. Display the player’s current level and score.

Hint: A player is considered alive if the health value is greater than 0.

8. Train Seat Status

Create a class named `TrainSeat` containing the following private data members:

- Seat Number
- Passenger Name
- Booking Status

Create a class named `TicketChecker` and declare it as a **friend class** of `TrainSeat`.

The `TicketChecker` class should:

1. Display the seat details.
2. Check whether the seat is booked or available.
3. Display the passenger name if the seat is booked.

Hint: The friend class can directly access the private members of `TrainSeat`.

9. Online Exam Result

Create a class named `Exam` containing the following private data members:

- Student Name
- Subject
- Marks
- Maximum Marks

Create a class named `Result` and declare it as a **friend class** of `Exam`.

The `Result` class should:

1. Access the private members of `Exam`.
2. Calculate the percentage.
3. Display "Pass" if the percentage is 40% or above; otherwise display "Fail".
4. Display the complete result.

Hint: $Percentage = \frac{Marks}{Maximum\ Marks} \times 100$.

10. Smart Home Device

Create a class named `SmartDevice` containing the following private data members:

- Device Name
- Device Type
- Power Status

Create a class named `HomeController` and declare it as a **friend class** of `SmartDevice`.

The `HomeController` class should provide member functions to:

1. Display device information.
2. Turn the device ON.
3. Turn the device OFF.
4. Display the current power status.

Hint: Since `HomeController` is a friend class, its member functions can directly access and modify the private members of `SmartDevice`.